

Tarun Mittal

Senior Software Engineer

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SUMMARY

- Senior Software Engineer with **5 years** of experience designing and owning cloud-native, high-throughput **distributed systems**.
- Expertise in **Python and Go microservices**, with applied experience in **AI-assisted and RAG systems**.
- Proven track record shipping production systems, leading backend migrations, and improving reliability, performance, and scalability.

SKILLS

- Languages** – Python, Go (Golang), SQL, C, C++.
- Backend & APIs** – FastAPI, Gin, Django, Pydantic, REST, gRPC, Celery, asyncio.
- AI & Agents** – LLM agents (OpenAI/Gemini/Claude), LangGraph, LangChain, MCP, tool calling, prompt engineering, RAG, embeddings, Milvus, pgvector, offline eval, LangSmith.
- Streaming & Data** – Apache Kafka, Apache Flink, ClickHouse, BigQuery, Batch Processing, ETL.
- Databases & Storage** – PostgreSQL, MySQL, Redis, MongoDB, Elasticsearch, DynamoDB, S3, Caching, Sharding.
- Cloud & Infrastructure** – GCP, AWS, Docker, Kubernetes, CI/CD, Bazel, Linux, New Relic, Datadog, PostHog, LangSmith.
- Core Engineering** – Distributed Systems, System Design (HLD/LLD), Multithreading, Concurrency, Caching, Sharding, Reliability, Testing.

PROFESSIONAL EXPERIENCE

Senior Software Engineer IMPACT ANALYTICS, Bangalore May 2026 – Present

Agentic AssortSmart (AI-Powered Retail Merchandise Planning)

- Building **AssortSmart**, a retail merchandise planning platform for seasonal buying, store clustering, and assortment decisions.
- Architected AssortSmart's **Keep/Drop engine** at article $\times$ plan-season grain, combining deterministic **ST%/ROS** scoring with **LangGraph** lenses, kept agents **SELECT-only** on ClickHouse through CSV-first bake-and-promote, with promotions gated on **300 gold cases** and **$\geq 80\%$ offline accuracy**.
- Built a **read-only dig-deeper QnA agent** over locked Keep/Drop decisions, enabling planners to understand why styles were kept or dropped while schema constraints preserved frozen decisions and blocked writes to ClickHouse, CSVs, and outcomes.
- Drove adoption of **ClickHouse** as AssortSmart's planning analytics engine, reducing pivot latency from **189s to 12.3s ($\sim 15.5x$)** on **250M rows** through a row-identical Postgres-versus-ClickHouse POC.
- Tech** – Go, Gin, Python, FastAPI, LangGraph, MCP, ClickHouse, BigQuery, GCS, Datadog, LangSmith, PostHog, GCP, Docker.

Software Development Engineer 2

UBER (via EPAM Systems), Bangalore

July 2024 – May 2026

Financial Risk Management (FRM) Scoping Platform (Uber Finance)

- Owned Uber's **FRM Risk Scoping backend (FastAPI, MySQL)** across **30+ REST APIs** powering **8** scoping screens, automating quarterly close at **\$340M group materiality** and targeting a **70% cut** in manual reconciliation, from about **2 weeks to 3–4 days**.
- Owned the Sheets $\rightarrow$ **MySQL** recon v2 migration (**18 files**), shipping parallel **v2 APIs** that build **L1 $\rightarrow$ L2 $\rightarrow$ L3** FSLI trees, tie **HFM** amounts to public **10-Q** filings, and give each line a durable ID for audit.
- Automated in-scope decisions across **~ 55 line items $\times$ 14 entities** by encoding materiality, qualitative override, residual-risk, and **5%** component-threshold logic in the scoping services against **\$340M / \$170M** thresholds.
- Led 3 engineers** on the FRM backend by owning API contracts and layered design reviews for Finance's quarterly scoping.
- Tech** – Python, FastAPI, SQLAlchemy 2.0, Pydantic, MySQL, Bazel, Docker.

Menu Ingestion & Automation Platform (Uber Eats)

- Owned Uber Eats menu ingestion end to end on **GCP**, landing partner menus through **Selenium** acquisition, a **Kafka** ingest bus, and **Flink** keyed normalize/dedupe — cutting onboarding from **24 hours to 2 hours** and saving **\$600K+/yr** at **30K+ menus/month**.
- Turned multilingual **PDF/image** menus into Uber Eats catalog schema at **98% fidelity / 100% schema consistency** (offline eval) by owning a **LangChain RAG + Gemini 2.5 Pro** path over **Milvus** embeddings with a hard schema validation gate before upsert.
- Raised successful menu ingestions to **95%+** by hardening the scrape fleet with IP rotation, dynamic proxy pools, and adaptive retries.
- Tech** – Python, Selenium, Kafka, Flink, LangChain, Gemini, RAG, Milvus, GCP, Docker.

ANZ Driver Document Compliance (Uber Mobility)

- Automated Uber Mobility driver and vehicle document checks for **ANZ** against local authority requirements, reaching **99.9%** compliance and removing **~ 20 hours/week** of manual verification.

Software Development Engineer 2

MASTERS INDIA, Noida

December 2022 – June 2024

GST Compliance & E-Invoicing Platform

- Migrated a PHP (**Laravel**) monolith to **FastAPI microservices** for **1,500+ clients**, bringing p95 from **1.2s to 300ms** while mentoring **2 engineers**.
- Built a bulk e-invoice path on **Kafka** and **PostgreSQL** (tables split by tax quarter) that processed **1M+ government submissions/day** and **100K+/import**, lifting throughput from **700 to 4,000 requests/min**.
- Made IRP retries safe with **idempotency keys, retries, and a dead-letter queue**, and used Redis caching to lower repeat DB reads by **30%**.
- Set up **ELK and New Relic** alerting plus a client usage dashboard that lowered incident triage **70%** and support tickets **35%**, raising coverage **35% to 82%** at **98%** deploy success.
- Tech** – Python, FastAPI, Kafka, PostgreSQL, MongoDB, Redis, Elasticsearch, Celery, Docker, ELK, New Relic, AWS.

Software Development Engineer

GEEKSFORGEEKS, Noida

August 2021 – November 2022

- Stabilized doubt-support for **10K+ daily queries** and **10x** contest spikes by migrating the backend from PHP to **Django (Python)**.
- Lifted premium subscriptions **15–20%** by shipping voting, pinning, and locking REST APIs on MySQL, MongoDB, Redis, and Elasticsearch.
- Raised course sales **30%** by launching an influencer earnings and analytics dashboard (transactions, coupons, filters).
- Raised operational efficiency **70%** with scheduled jobs for video processing, reminders, and recording cleanup.

EDUCATION

B.Tech (Information Technology)

Institute of Engineering & Technology, Lucknow

July 2017 – June 2021

ACHIEVEMENTS & CERTIFICATIONS

- Ranked **2260 / 37,000+** in the Google Code Jam Qualifier 2021 and finalist (top 3 nationally) at the **Smart India Hackathon 2020 (MHRD)**.
- Participated in the **Global AI Hackathon** at EPAM Systems.
- Earned **HackerRank Problem Solving** and **LangChain Academy** certifications ([HackerRank](#), [LangChain](#)).